

Interview Report Summary

1. Stakeholder Engagement Status: The team has successfully engaged with three key stakeholder categories to validate the problem of reactive fire management in Ourense.
 - *Rafael (Forest Agent, 42)*: Completed (Nov 15). Provided technical field insights on the need for dynamic vs. static data.
 - *Maria (Local Council Officer)*: Completed (Nov 19). Highlighted the communication gap between technical data and public safety announcements.
 - *José (Local Neighbor/Landowner, 63)*: Completed (Dec 21). Provide the citizen perspective on asset protection and community anxiety.
 - *Manuel (President of “Comunidad de Montes”)*: Completed (Jan 12). Represents the communal landowners who manage the biomass.
 - *Climate Data Analyst*: Pending Confirmation. Outreach initiated to a specialist at MeteoGalicía to validate our data sourcing variables.
2. Key Insights from Interviews (Synthesis): The addition of the Comunero perspective has shifted our understanding from just “emergency response” to “budget optimization”.
 - **The Budget Lottery (Manuel – Comunidad de Montes)**: Manuel revealed that his community manages 500 hectares but only has the budget to clear (brush-cut) 20 hectares per year. Currently, they choose these zones based on intuition. He stated, “We are guessing. If your AI tells me which 20 hectares are most likely to ignite, I will spend my budget there.” This confirms a B2B business model for preventive planning.
 - **The Static Map Frustration (Rafael & Maria)**: Confirmed that static paper maps are obsolete by the time they are distributed. Real-time updates are non-negotiable.
 - **The Passive Observer Anxiety (José)**: neighbors feel helpless not about their own land, but about the unknown risks of adjacent abandoned properties.

Technical Literature Critique

Selected Paper: Piao, Y., Lee, D., Park, S., Kim, H. G., & Jin, Y. (2022). Forest fire susceptibility assessment using google earth engine in Gangwon-do, Republic of Korea. *Geomatics, Natural Hazards and Risk*, 13(1), 432–450.
<https://doi.org/10.1080/19475705.2022.2030808>

1. **Relevance to Project.** This paper mirrors our technical architecture. The authors built a Forest Fire Susceptibility Map (FFSM) using Google Earth Engine (GEE) and a Random Forest algorithm. They used variables identical to our roadmap: slope (topography), NDVI (vegetation), and climate data to predict risk zones.

2. Critique & Reflection

- Strengths (Validation):
 - *Algorithm Choice:* The study achieved an AUC of 0.835 with Random Forest, validating our decision to use this over simpler regression models for handling complex, non-linear variables like biomass density.
 - *Cloud Capability:* It proves Google Earth Engine can process multi-spectral Sentinel-2 data combined with topographic data (DEM) without heavy local infrastructure, confirming our “Cloud Processing” strategy is viable and scalable.
- Weaknesses (Gaps):
 - *Temporal Resolution:* The paper focuses on long-term susceptibility (where fires might happen generally) rather than dynamic daily prediction. Our project promises “Real-time data processing” and “Hyperlocal predictions for this afternoon”. The paper’s model is too static for our user needs.
 - *Lack of Thermal Input:* The paper did not utilize real-time thermal anomaly data (Sentinel-3 /MODIS). Our solution explicitly includes “Heat accumulation” as a variable, which improves upon this paper’s methodology by adding immediate pre-ignition detection.
- Adaptability:
 - The GEE code is highly adaptable. We can swap their “Gangwon-do” region for “Ourense” but must modify the output from a probability percentage to our proposed “Traffic Light” visualization for simpler user adoption.

Appendix

A. Interview Transcript Snippet (Translated into English): Manuel (Comunidad de Montes President) *Date: Jan 12, 2026* | Topic: Preventive Budgeting

- **Marc:** “How do you decide where to send the brush-cutters each spring?”
- **Manuel:** “It’s a discussion. One neighbor says the north slope is overgrown, another says the path to the shrine is blocked. We have money to clean maybe 5% of the land. It’s a lottery.”
- **Marc:** “If we gave you a map showing which areas have the driest biomass and highest fuel load, would you use it?”
- **Manuel:** “If it works, yes. It justifies the expense to the assembly. I can say, “We are clearing Zone B because the machine says it will burn, not just because I said so.”

B. Data Layer Dictionary

Based on Technical Approach and Literature Review.

Data Source	Variable	Project Purpose
Sentinel-2	NDVI / Moisture	Biomass density & vegetation health
Sentinel-3 /MODIS	LST (Land Surface Temp)	Heat accumulation (Real-time thermal)
DEM (SRTM)	Slope / Aspect	Fire spread speed (Topography)
AEMET (API)	Wind / Humidity	Weather forcing (Hyperlocal)